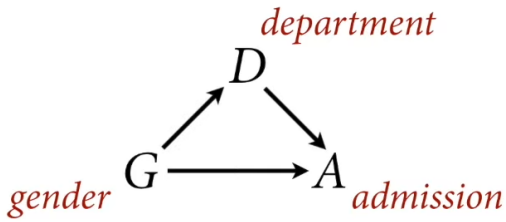
Lecture 33: GLMs in practice on examples

A simulated model of the University admission framework with discrimination

```
# generative model, basic mediator scenario

N <- 1000 # number of applicants
# even gender distribution
G <- sample( 1:2 , size=N , replace=TRUE )
# gender 1 tends to apply to department 1, 2 to 2
D <- rbern( N , ifelse( G==1 , 0.3 , 0.8 ) ) + 1
# matrix of acceptance rates [dept,gender]
accept_rate <- matrix( c(0.05,0.2,0.1,0.3) , nrow=2 )
# simulate acceptance
A <- rbern( N , accept_rate[D,G] )
```



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Admission simulation discrimination

Two models

Total Effect

$$
\begin{aligned}
A_i &\sim \text{Bernoulli}(p_i) \\
\text{logit}(p_i) &= \alpha[G_i] \\
\alpha_j &\sim \text{Normal}(0, 1)
\end{aligned}
$$

Direct Effect(s)

$$
\begin{aligned}
A_i &\sim \text{Bernoulli}(p_i) \\
\text{logit}(p_i) &= \alpha[G_i, D_i] \\
\alpha_{j,k} &\sim \text{Normal}(0, 1)
\end{aligned}
$$

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Admission simulation discrimination

Two models

Total Effect

```
dat_sim <- list( A=A , D=D , G=G )

m1 <- ulam(
    alist(
        A ~ bernoulli(p),
        logit(p) <- a[G],
        a[G] ~ normal(0,1)
    ), data=dat_sim , chains=4 , cores=4 )
```

```
precis(m1,depth=2)
```

	mean	sd	5.5%	94.5%	n_eff	Rhat4
a[1]	-1.80	0.13	-2.01	-1.60	1549	1
a[2]	-1.09	0.10	-1.25	-0.93	1159	1

Direct Effect(s)

```
dat_sim <- list( A=A , D=D , G=G )

m2 <- ulam(
    alist(
        A ~ bernoulli(p),
        logit(p) <- a[G,D],
        matrix[G,D]:a ~ normal(0,1)
    ), data=dat_sim , chains=4 , cores=4 )
```

```
precis(m2,depth=3)
```

	mean	sd	5.5%	94.5%	n_eff	Rhat4
a[1,1]	-2.31	0.18	-2.60	-2.04	2529	1
a[1,2]	-0.92	0.19	-1.23	-0.62	2216	1
a[2,1]	-1.93	0.31	-2.45	-1.44	2214	1
a[2,2]	-0.93	0.11	-1.11	-0.75	2055	1

```
> inv_logit(coef(m2))
    a[1,1]     a[1,2]     a[2,1]     a[2,2]
0.06296434 0.21109945 0.08253890 0.20003819
```

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Analyse the UC Berkeley Admissions

- 4526 graduate school applications for 1973 UC Berkeley
- Stratified by department and gender of applicant
- See `?UCBAdmissions` for details and citations

- Evidence of gender discrimination?

	dept	admit	reject	applications	gender
1	A	512	313	825	male
2	A	89	19	108	female
3	B	353	207	560	male
4	B	17	8	25	female
5	C	120	205	325	male
6	C	202	391	593	female
7	D	138	279	417	male
8	D	131	244	375	female
9	E	53	138	191	male
10	E	94	299	393	female
11	F	22	351	373	male
12	F	24	317	341	female

Logistic or Binomial regression

Six of one and half a dozen of the other

Logistic regression: Binary { 0,1 } outcome and logit link

$$\begin{aligned} A_i &\sim \text{Bernoulli}(p_i) \\ \text{logit}(p_i) &= \alpha[G_i, D_i] \end{aligned}$$

Binomial regression: Count { 0, 1, . . . , n } outcome and logit link

$$\begin{aligned} A_i &\sim \text{Binomial}(n_i, p_i) \\ \text{logit}(p_i) &= \alpha[G_i, D_i] \end{aligned}$$

Both approaches are completely equivalent for inference

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Logistic or Binomial regression

Six of one and half a dozen of the other

LOOOOOOOONG

	A	G	D
1	0	2	1
2	0	1	1
3	1	2	2
4	1	2	2
5	0	2	2
6	0	1	1
7	0	2	2
8	0	2	2
9	0	2	2
10	0	2	2
11	0	2	2
12	1	2	2
13	0	2	2
14	0	1	1
15	0	2	2
16	0	1	2
17	0	1	1
18	0	1	1
19	0	1	1
20	0	1	1

```
dat_sim2 <- aggregate( A ~ G + D , dat_sim , sum )
dat_sim2$N <- aggregate( A ~ G + D , dat_sim , length )$A
```

	G	D	A	N
1	1	1	30	355
2	2	1	10	92
3	1	2	38	135
4	2	2	117	418

Aggregated

Binomial regression

for the UC Berkeley admission data

```
m2 <- ulam(
    alist(
        A ~ bernoulli(p),
        logit(p) <- a[G,D],
        matrix[G,D]:a ~ normal(0,1)
    ), data=dat_sim , chains=4 , cores=4 )
```

$$
\begin{aligned}
A_i &\sim \text{Bernoulli}(p_i) \\
\text{logit}(p_i) &= \alpha[G_i, D_i]
\end{aligned}
$$

```
m2_bin <- ulam(
    alist(
        A ~ binomial(N,p),
        logit(p) <- a[G,D],
        matrix[G,D]:a ~ normal(0,1)
    ), data=dat_sim2 , chains=4 , cores=4 )
```

$$
\begin{aligned}
A_i &\sim \text{Binomial}(n_i, p_i) \\
\text{logit}(p_i) &= \alpha[G_i, D_i]
\end{aligned}
$$

Both approaches are completely equivalent for inference

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UC Berkeley Admission analysES

Total Effect

```
data(UCBadmit)
d <- UCBadmit

dat <- list(
    A = d$admit,
    N = d$applications,
    G = ifelse(d$applicant.gender=="female",1,2),
    D = as.integer(d$dept)
)

# total effect gender
mG <- ulam(
    alist(
        A ~ binomial(N,p),
        logit(p) <- a[G],
        a[G] ~ normal(0,1)
    ), data=dat , chains=4 , cores=4 )
```

Direct Effect(s)

```
# direct effects
mGD <- ulam(
    alist(
        A ~ binomial(N,p),
        logit(p) <- a[G,D],
        matrix[G,D]:a ~ normal(0,1)
    ), data=dat , chains=4 , cores=4 )
```

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UC Berkeley Admission data analyses outputs

OMG

```
# total effect gender
mG <- ulam(
    alist(
        A ~ binomial(N,p),
        logit(p) <- a[G],
        a[G] ~ normal(0,1)
    ), data=dat , chains=4 , cores=4 )
```

```
# direct effects
mGD <- ulam(
    alist(
        A ~ binomial(N,p),
        logit(p) <- a[G,D],
        matrix[G,D]:a ~ normal(0,1)
    ), data=dat , chains=4 , cores=4 )
```

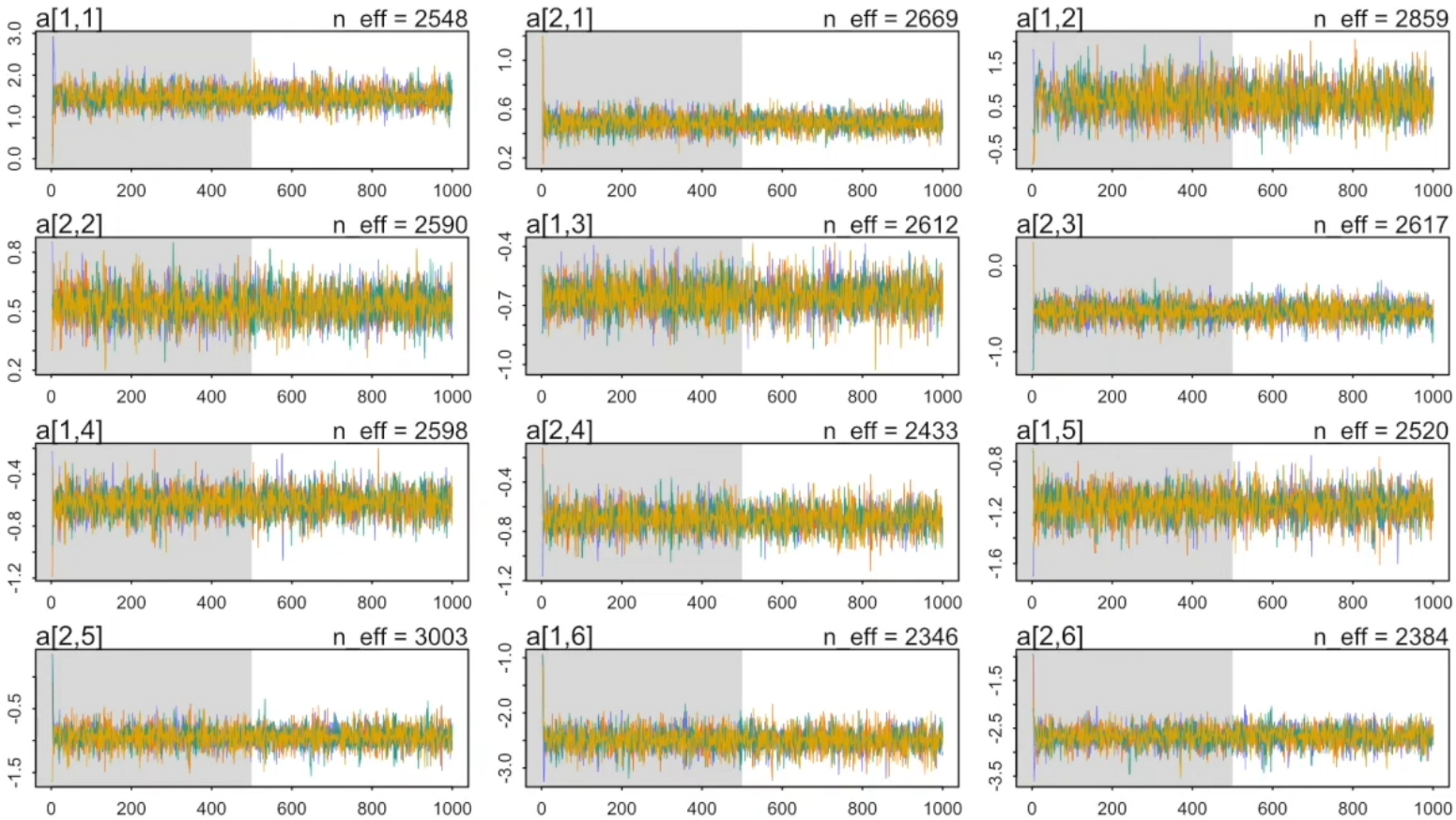
	mean	sd	5.5%	94.5%	n_eff	Rhat4
a[1]	-0.83	0.05	-0.91	-0.75	1487	1
a[2]	-0.22	0.04	-0.28	-0.16	1499	1

	mean	sd	5.5%	94.5%	n_eff	Rhat4
a[1,1]	1.48	0.24	1.11	1.87	2548	1
a[1,2]	0.66	0.40	0.03	1.32	2859	1
a[1,3]	-0.65	0.08	-0.79	-0.52	2612	1
a[1,4]	-0.62	0.11	-0.79	-0.45	2598	1
a[1,5]	-1.15	0.12	-1.34	-0.96	2520	1
a[1,6]	-2.50	0.20	-2.81	-2.18	2346	1
a[2,1]	0.49	0.07	0.38	0.60	2669	1
a[2,2]	0.53	0.08	0.40	0.67	2590	1
a[2,3]	-0.53	0.11	-0.72	-0.35	2617	1
a[2,4]	-0.70	0.10	-0.87	-0.54	2433	1
a[2,5]	-0.94	0.16	-1.20	-0.69	3003	1
a[2,6]	-2.67	0.21	-3.00	-2.34	2384	1

Well the diagnostics show green light all the way.

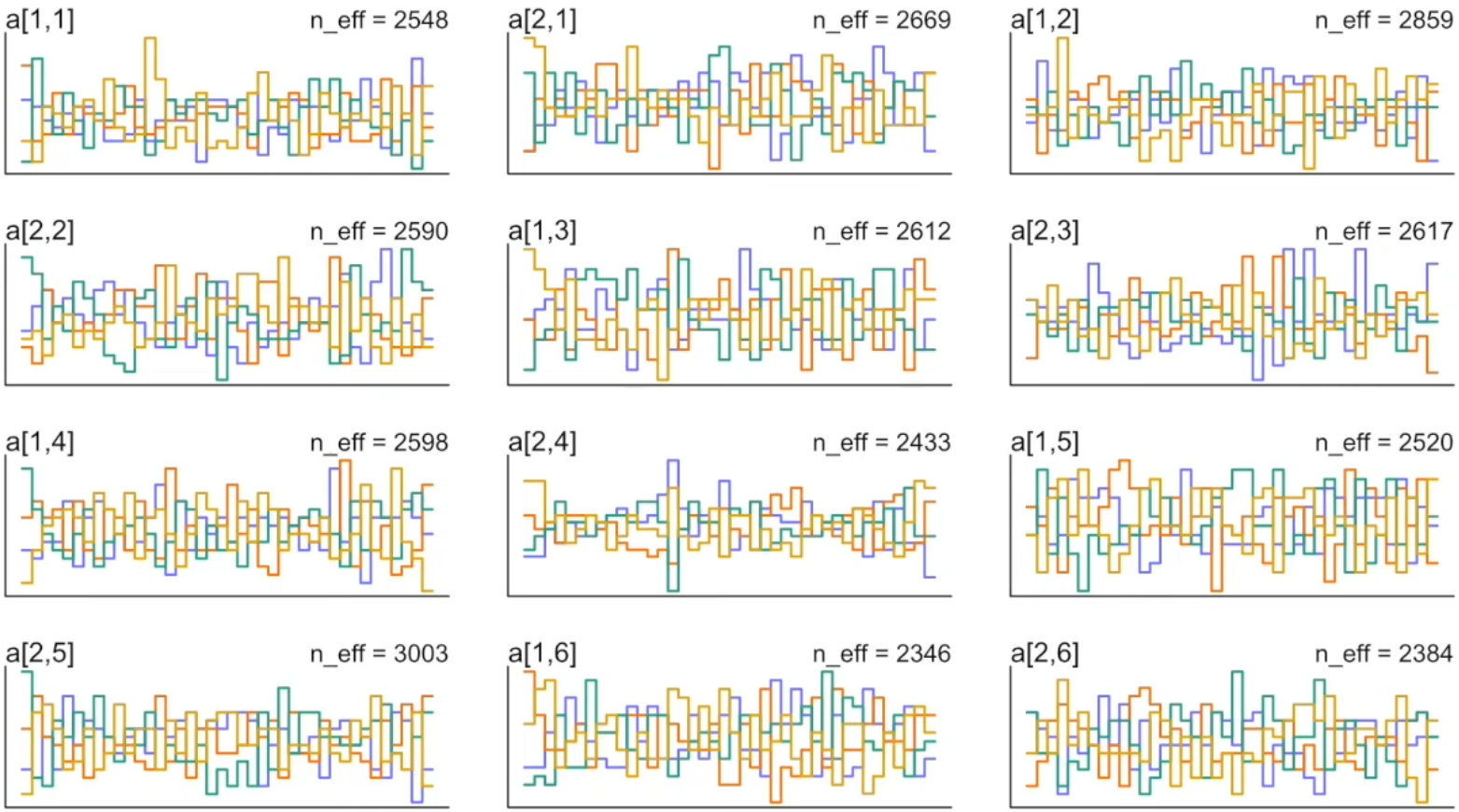
UC Berkeley Admission data analyses outputs

Trace plots



UC Berkeley Admission data analyses outputs

Trank plots



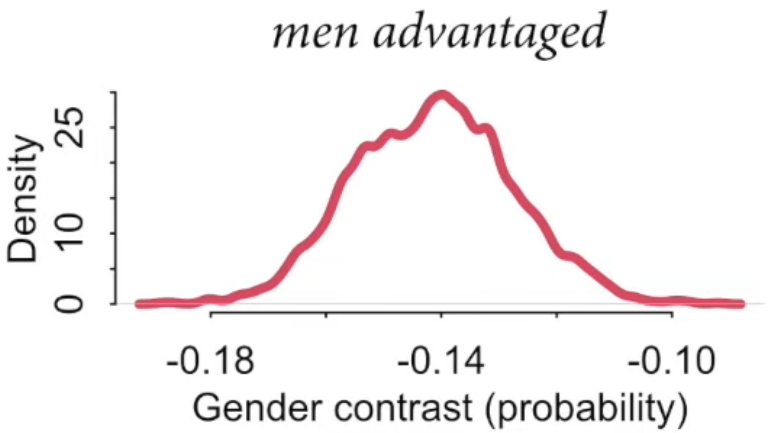
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UC Berkeley Admission data analyses outputs

Let's have a look at contrasts

Total Effect

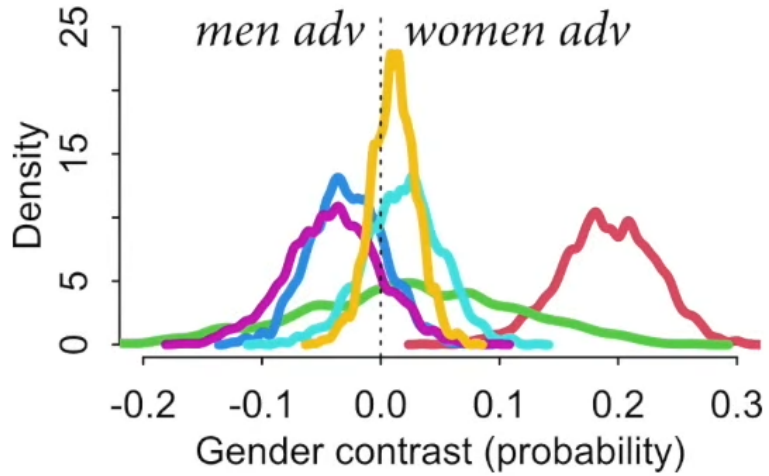
```
post1 <- extract.samples(mG)
PrA_G1 <- inv_logit( post1$a[,1] )
PrA_G2 <- inv_logit( post1$a[,2] )
diff_prob <- PrA_G1 - PrA_G2
dens(diff_prob,lwd=4,col=2,xlab="Gender
contrast (probability)")
```



Direct Effect(s)

```
post2 <- extract.samples(mGD)
PrA <- inv_logit( post2$a )
diff_prob_D_ <- sapply( 1:6 , function(i) PrA[,
1,i] - PrA[,2,i] )

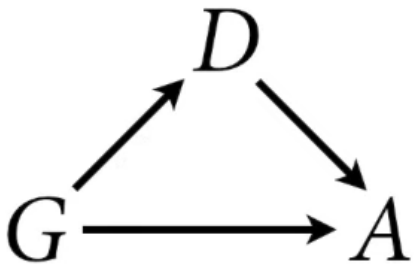
plot(NULL,xlim=c(-0.2,0.3),ylim=c(0,25),xlab="G
ender contrast (probability)",ylab="Density")
for ( i in 1:6 ) dens( diff_prob_D_[,i] , lwd=4
, col=1+i , add=TRUE )
```



UC Berkeley Admission data analyses outputs

- We estimate the direct effect in each department of hypothetically intervening on the application gender.
- What is the actual *average direct effect* of gender across departments?
- It depends upon the distribution of applications across departments, the probability of woman/man who applies to each department

What is the intervention actually?

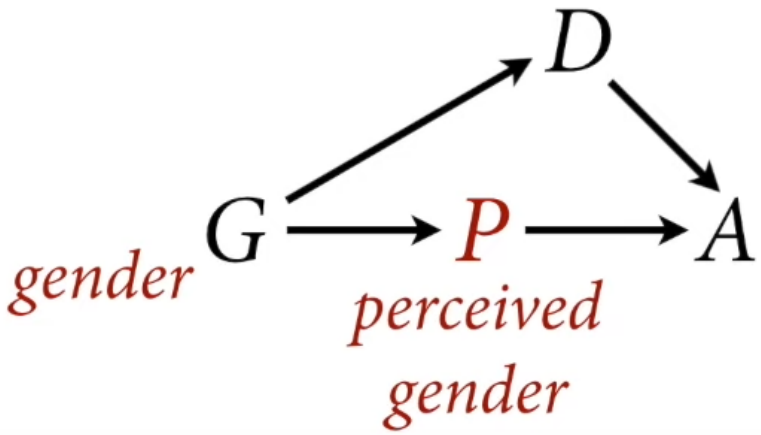


UC Berkeley Admission data analyses outputs

Average direct effect across departments?

- We estimate the direct effect in each department of hypothetically intervening on the application gender. We are actually changing the perceived gender $\rightarrow$ modified DAG
- What is the actual *average direct effect* of gender across departments?
- It depends upon the distribution of applications across departments, the probability of woman/man who applies to each department

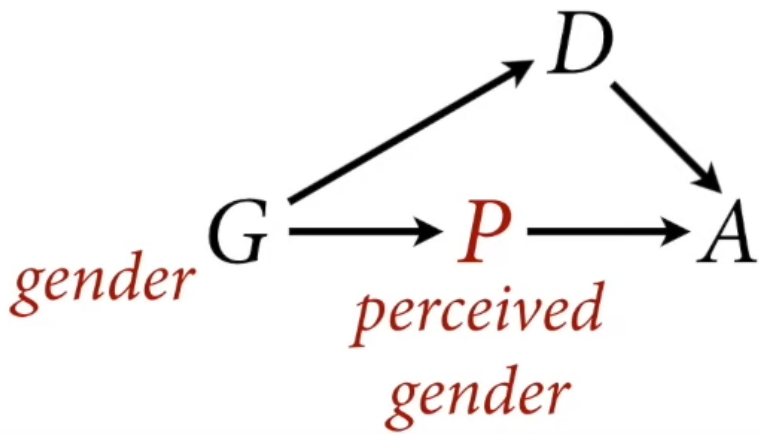
What is the intervention actually?



UC Berkeley Admission data analyses outputs

Average direct effect across departments?

What is the intervention actually?



- To calculate the causal effect of the mediator P, we must average (marginalise) over departments
- Easy to do as a simulation; as usual, easier to think about, easier to modify

UC Berkeley Admission data analyses

Simulating consequences of $M \rightarrow F$ gender intervention average effect

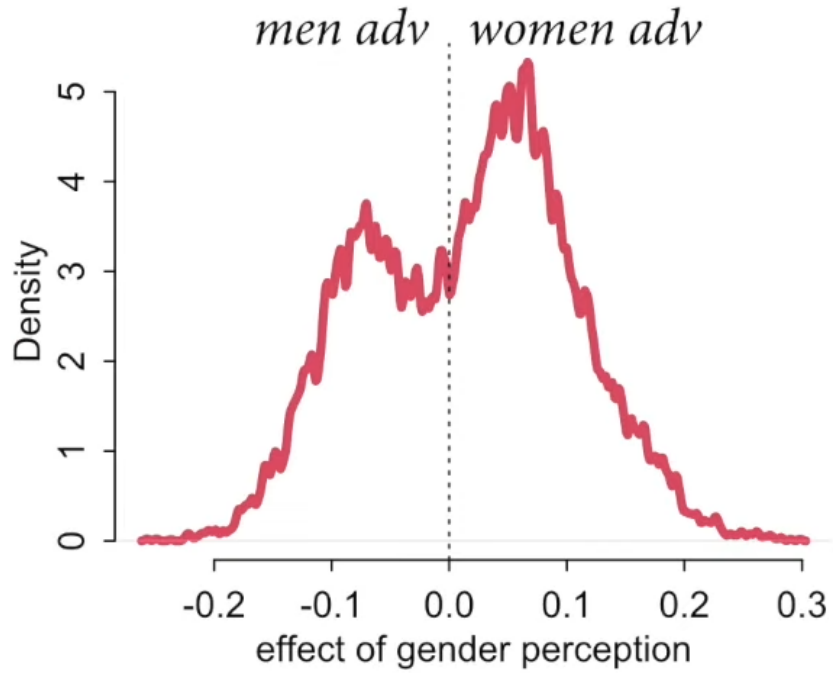
```
# number of applicatons to simulate
total_apps <- sum(dat$N)

# number of applications per department
apps_per_dept <- sapply( 1:6 , function(i)
sum(dat$N[dat$D==i]) )

# simulate as if all apps from women
p_G1 <- link(m2,data=list(
    D=rep(1:6,times=apps_per_dept),
    N=rep(1,total_apps),
    G=rep(1,total_apps)))

# simulate as if all apps from men
p_G2 <- link(m2,data=list(
    D=rep(1:6,times=apps_per_dept),
    N=rep(1,total_apps),
    G=rep(2,total_apps)))

# summarize
dens( p_G1 - p_G2 , lwd=4 , col=2 ,
xlab="effect of gender perception" )
```



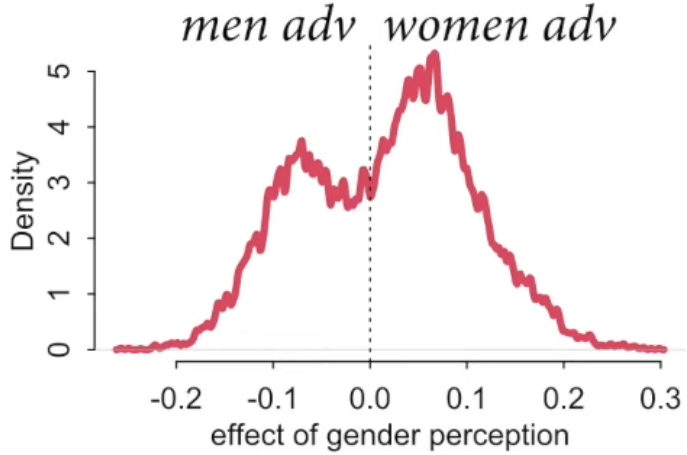
Peak is mainly due to largest department A where women have an advantage. Mean is not very representative.

UC Berkeley Admission data analyses

Simulating consequences of $M \rightarrow F$ gender intervention

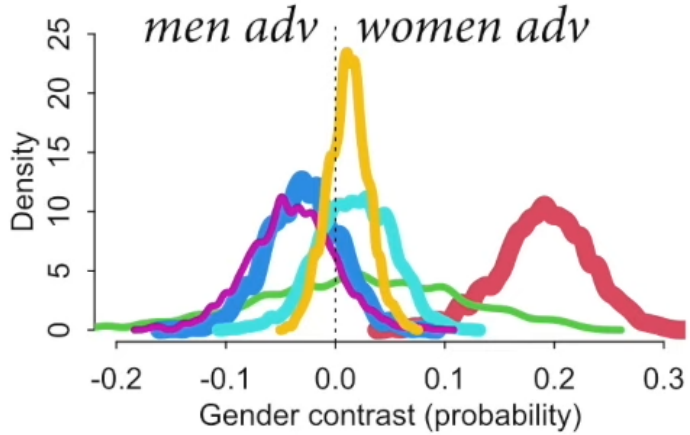
```
# simulate as if all apps from women
p_G1 <- link(m2,data=list(
    D=rep(1:6,times=apps_per_dept),
    N=rep(1,total_apps),
    G=rep(1,total_apps)))

# simulate as if all apps from men
p_G2 <- link(m2,data=list(
    D=rep(1:6,times=apps_per_dept),
    N=rep(1,total_apps),
    G=rep(2,total_apps)))
```




```
# show each dept density with weight as in
population
w <- xtabs( dat$N ~ dat$D ) / sum(dat$N)

plot(NULL,xlim=c(-0.2,0.3),ylim=c(0,25),xlab="
Gender contrast (probability)",ylab="Density")
for ( i in 1:6 ) dens( diff_prob_D_[,i] ,
lwd=2+20*w[i] , col=1+i , add=TRUE )
abline(v=0,lty=3)
```

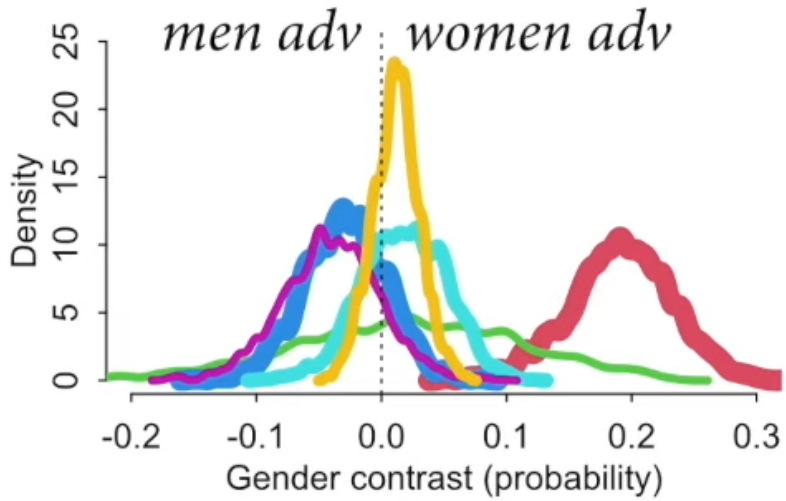


Department line thickness representative of #(applications)

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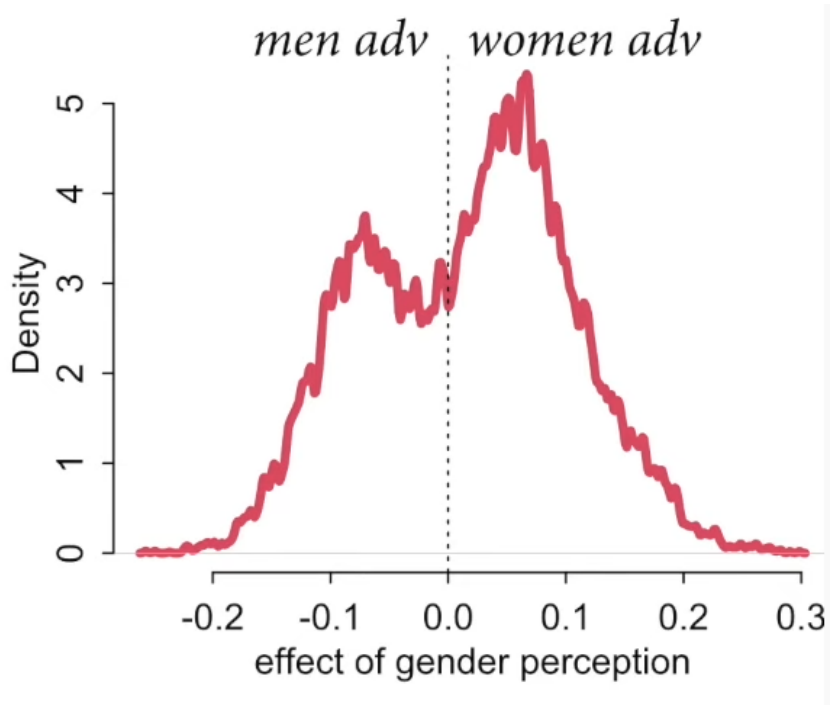
Post-stratification

- Description, prediction and causal inference often require *post-stratification*
- Post-stratification: re-weighting estimates for target population
- At different universities, the distributions of applications will differ; the predicted consequences of intervention will differ



UC Berkeley Admission data analyses so far

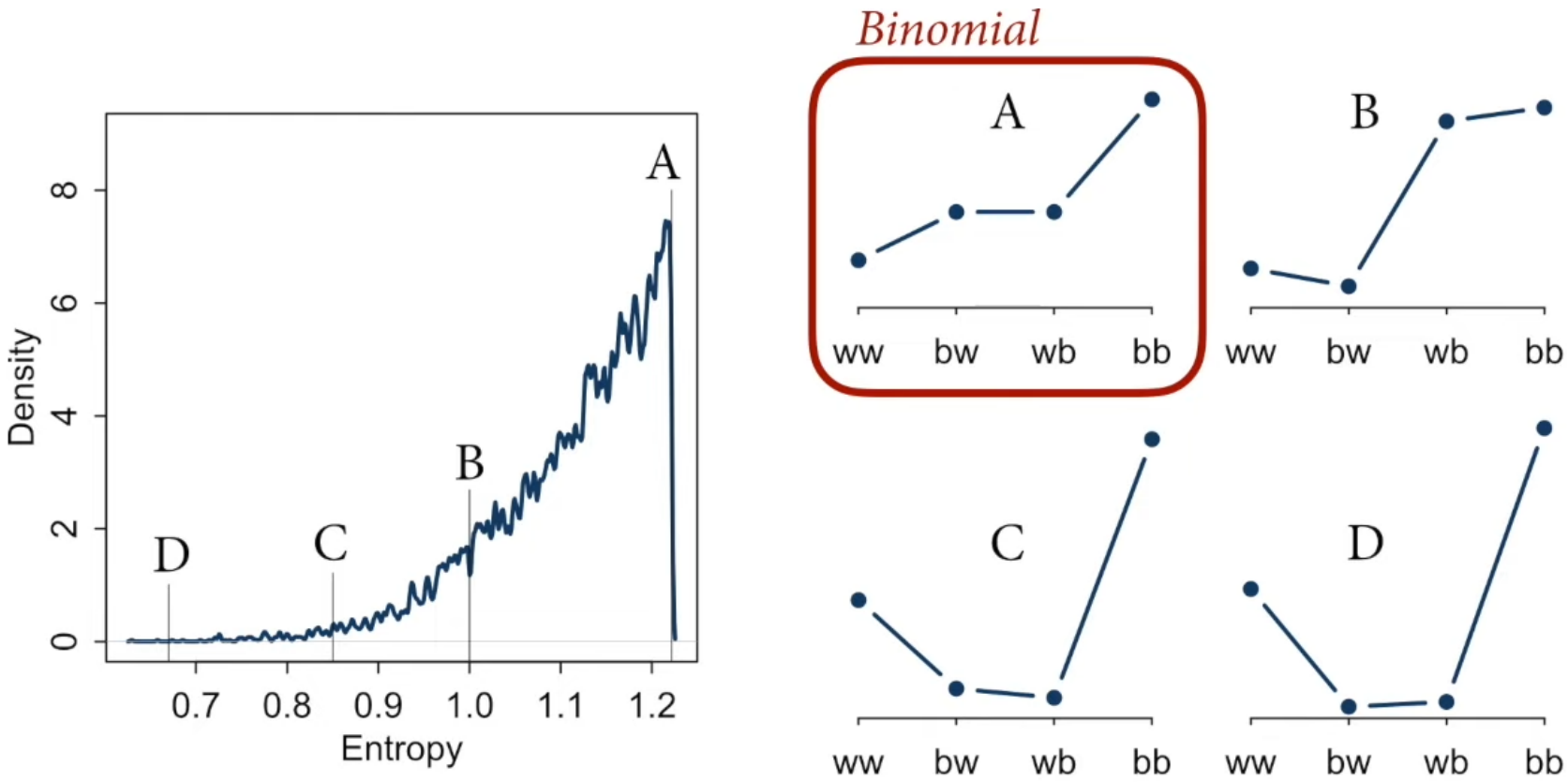
- Evidence for discrimination?
- No evidence so far, at least not direct (balance in plot on the right) but:
  - Distribution of applications can be a consequence of discrimination; data do not speak of this: negative cultural environment or fear of perceived discrimination Doesn't mean it's not relevant.
  - Confounds are likely



Summary of Binomial GLMs

- Outcome is a count with a maximum
- Maximum doesn't need to be known. Sometimes, it is even the target of the inference: population size estimation.
- logit link is not the only option
- The binomial distribution is the maximum entropy distribution for counts with constant expectation.

Binomial distribution is the unique "flattest" among all distributions with constant expectation



(Figure 10.4, SR book)

GLMs

A few additional remarks 1

- Omitted variable bias. We saw examples of such biases (Waffle House data, Milk energy/ńeocortex percent data)
  Worse with GLMs. Even if the variable isn't technically a confounder, the inference can be biased because of the floor/ceiling effects of the link function. The causal influence of a variable can be suppressed. See small example p320, SR book.
- Each parameter represents a relative difference on the linear model scale, ignoring other parameters. We are actually interested in the absolute differences in outcomes that need incorporate all parameters.

GLMs

A few additional remarks 2

- Information criteria are based on deviance, so their magnitude depends on constants that are distribution specific. Moral: only compare models with AIC/WAIC/PSIS that rely on the same type of likelihood (luckily, maxent can help in this choice). It is possible to compare models with different likelihood, just not with information criteria. Notice some nuances with WAIC/PSIS and the different GLMs.

- GLMs are easy to use with weakly informative priors. If prior knowledge dictates a shape, use it! You could also use maxent to choose priors.